

L13: Introduction to Network Sampling

It is often the case that we do not know the complete network. Instead, we only have sampled network data, and we need to rely on that sample to infer properties of the entire network. The area of *statistical sampling theory* provides some general approaches and results that are useful in this goal.

Let us denote the complete network as $G = (V, E)$ – we will be referring to it as the **population graph**.

Additionally, we have a **sampled graph** $G^* = (V^*, E^*)$, which consists of a subset of nodes V^* and edges E^* from the population graph.

To illustrate the challenges involved in network sampling, consider the following problem.

Suppose we want to estimate the average degree of the population graph, defined as

$$E[k] = \frac{\sum_{i \in V} k_i}{N}, \quad N = |V|$$

where k_i is the degree of node- i and N is the number of nodes in the population graph.

The obvious approach is to estimate this with the average degree of the sampled graph G^* as follows:

$$\bar{k} = \frac{\sum_{i \in V^*} k_i}{n}, \quad n = |V^*|$$

where n is the size of the sample.

Now, consider two different network sampling strategies, or “*designs*”. In both of them we start with a random sample V^* of n nodes:

Design-1: for each $i \in V^*$, copy all edges that are adjacent to node i in the set of sampled edges E^* .

Design-2: for each pair of sampled nodes $(i, j) \in V^* \times V^*$, examine if they are connected with an edge, i.e., if $(i, j) \in E$. If they are, copy that edge in E^* .

Note that Design-1 requires that we know *all neighbors of each sampled node*, even if those neighbors are *not* sampled – while Design-2 only observes the *adjacencies between sampled nodes*. In practice, this means that Design-2 would be simpler or less costly to conduct. Imagine, for example, that we construct a social network based on who is calling whom. Design-1 would require that we know all phone calls of the sampled individuals. Design-2 would require that we only know whether two sampled individuals have called each other.

Which of the two Designs will result in a better estimate of $E[k]$ in your opinion? Please think about this for a minute before you look at the answer below.

If you answered Design-1, you are right. With Design-2, we underestimate the degree of each node by a factor of roughly $\frac{n}{N}$ because, on the average, we only “see” that fraction of nodes of the population graph.

That does not mean however that Design-2 is useless. As we will see later in this Lesson, we could use Design-2 and get a reasonable estimate of the average degree as long as we add an appropriate “correction factor”. For example, we could get a better estimate with Design-2 if we use the following correction:

$$\bar{k} = \frac{N}{n} \frac{\sum_{i \in V^*} k_i}{n}$$

This can be a useful approach if Design-2 is simpler or cheaper to apply than Design-1.

Here is another example of how statistical sampling theory can be useful: imagine that you want to estimate the number N of nodes in the population network G . For practical reasons however, it is impossible to identify every single node in G -- you can only collect samples of nodes. How would you estimate N ? Please think about this for a minute before you read the following.

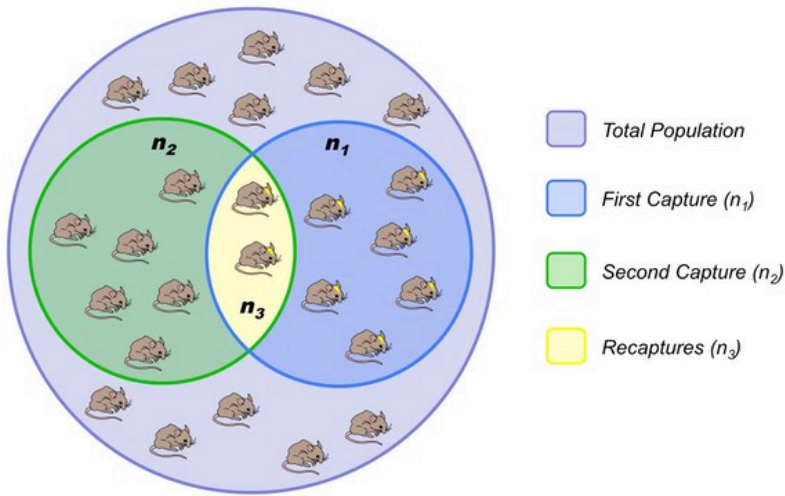


Image source: <http://www.old-ib.bioninja.com.au/options/option-g-ecology-and-conser/g5-population-ecology.html>

We could use a statistical technique called **capture-recapture estimation**. The simplest version is that we first select a random sample S_1 of n_1 nodes, without replacement. The important point is that we somehow “mark” these nodes. For instance, if they have a unique identifier, we record that identifier.

Then, we collect a second sample S_2 of n_2 nodes, again without replacement. Let n_3 be the number of nodes that appear in both S_1 and S_2 , i.e., $n_3 = |S_1 \cap S_2|$.

If $n_3 > 0$, we can estimate the size of the population graph as follows:

$$\hat{N} = \frac{n_1 n_2}{n_3}$$

What is the rationale behind this estimator? The probability that a node is sampled in both S_1 and S_2 can be estimated as:

$$\frac{n_3}{N} = \frac{n_1}{N} \frac{n_2}{N}$$

because the two samples are independently collected. Solving for N gives us the previous capture-recapture estimator.

In sampling theory, two important questions are:

- Is the estimator unbiased**, meaning that its expected value is equal to the population-level metric we try to estimate. In the previous example, is it that $E[\hat{N}] = N$, across all possible independent samples S_1 and S_2 ?
- What is the variance of the estimator?** Estimators with large variance are less valuable in practice.

For instance, it is not hard to show that the variance of the estimator $\hat{N}$ is:

$$V(\hat{N}) = \frac{n_1 n_2 (n_1 - n_3) (n_2 - n_3)}{n_3^3}$$

Suppose that $n_1 = n_2 = 1000$ and $n_3 = 1$. Our estimate of the population graph size will be $\hat{N} = 10^6$ – but that is also the value of the standard deviation of $\hat{N}$, making the estimate useless!

If $n_3 = 100$, however, we get that $\hat{N} = 10,000$ and the standard deviation of this estimate is 900, which is much more reasonable.

Food For Thought

For the previous estimator $\hat{N}$, show that if n_1 and n_2 are fixed, the variable n_3 follows the hypergeometric distribution. Use this result to derive the previous expression for the variance of $\hat{N}$.